

REDOX REACTION

Red - Reduction

OX - Oxidation

Definition:- The chemical Rxn in which both oxidation and Reduction are Simultaneously take Place. is known as Redox Reaction.

Oxidation State (O.S):-

O.S of an atom in ion or molecule is real or hypothetical charge Present on atom due to difference of electronegativity.

Some important Rule for Calculation of O.S:

① O.S of an element in Free state : "Zero"

Ex:- Na, Cl, Cu, Ne, He $\rightarrow$ O.S = Zero

② O.S of an atom in homoatomic Molecule is : "Zero"

Ex:- $H_2, N_2, Cl_2, O_2, P_4, S_8 \rightarrow$ O.S = Zero

③ O.S of an atom in Allotropic form is = "Zero"

$C_{\text{graphite}}, C_{\text{diamond}} \rightarrow$ O.S = "Zero"

④ O.S of all Component is in alloys = "Zero"

$Na - Hg \rightarrow$ O.S = Zero

⑤ In Complex Compound o.s of Some Neutral molecule is = "Zero"

Ex: NH_3 , H_2O , CO

⑥ O.S of Fluorine = -1 (always)

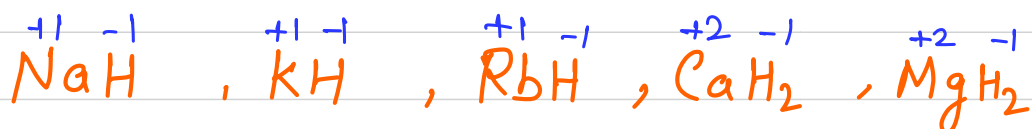
In General halogen (F, Cl, Br, I) = -1

⑦ Alkali Metal (Li, Na, K, Rb, Cs) IA group
O.S = +1 (always)

Alkaline earth Metals (Be, Mg, Ca, Sr, Ba)
IIA group $\rightarrow$ O.S = +2 (always)

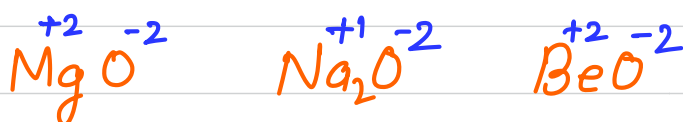
IIIA = Al O.S = +3

⑧ O.S of hydrogen Generally = +1
but in Case of Metal hydride O.S of Hydrogen = -1

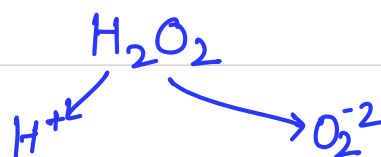
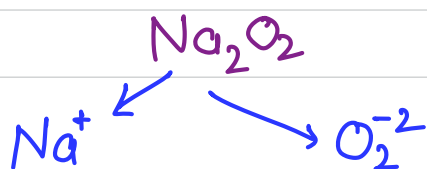


⑨ O.S of Oxygen Generally = -2

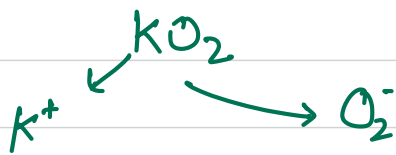
(a) oxide (O^{2-}) $\rightarrow$ O.S of O = -2



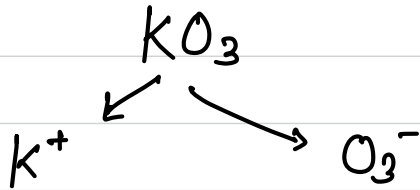
(b) Peroxide (O_2^{2-}) $\rightarrow$ O.S of O = -1



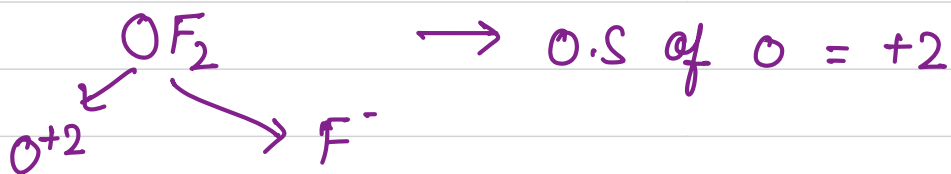
(c) - Superoxide (O_2^-) o.s of O = $-\frac{1}{2}$



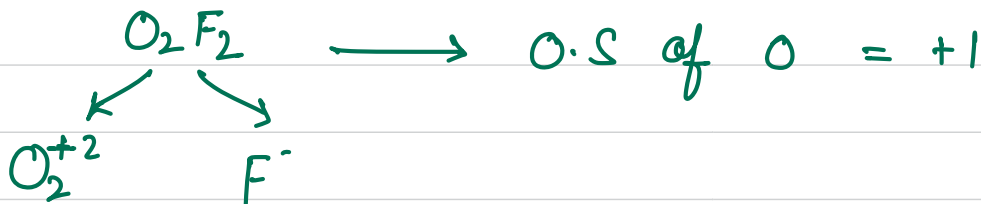
(d) Ozonide ion (O_3^-) $\rightarrow$ o.s of O = $-\frac{1}{3}$



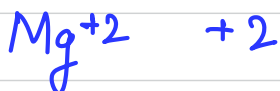
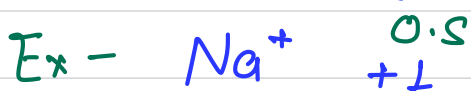
(e) Oxygen difluoride



(f) Dioxygen difluoride



(10) o.s of an element in monoatomic ion is equal to charge present on ion.



(11) Sum of o.s of all element Present in a Neutral molecule = Zero



$$(1 \times 2) + x + (-2 \times 4) = 0$$

$$2 + x - 8 = 0$$

$$x = +6$$

$$\underline{\underline{x = +6}}$$



$$2 + x - 6 = 0$$

$$x = +4$$

(12) Sum of O.S of all element in ion is equal to = charge present on ion.



$$x - 8 = -2$$

$$x = +6$$

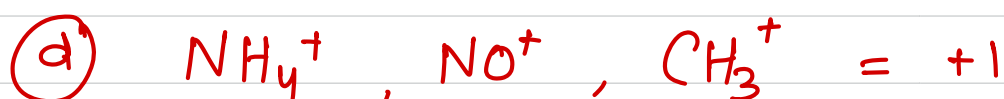
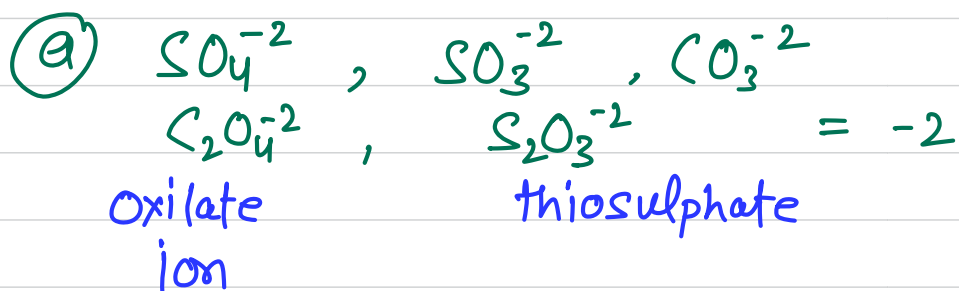


$$x - 6 = -2$$

$$x = +4$$

(13) O.S of an element Can be - Zero, +ve, -ve, fractional

(14) Charge on ion:

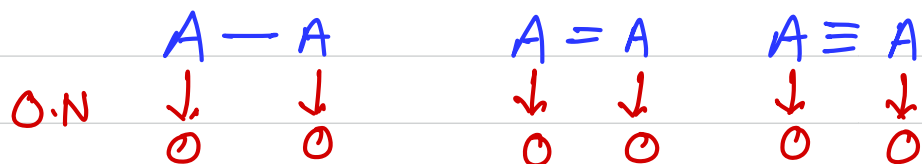


Calculation of O.N with the help of Structure:-

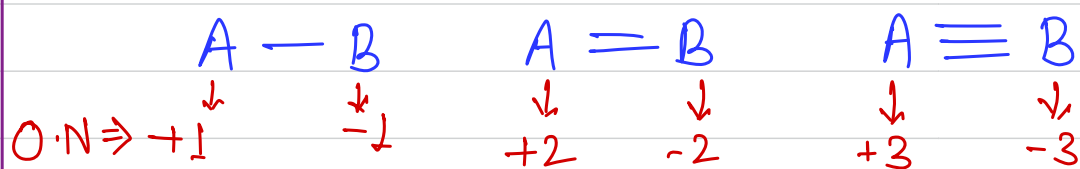
(A) In Case of Covalent Bond:-



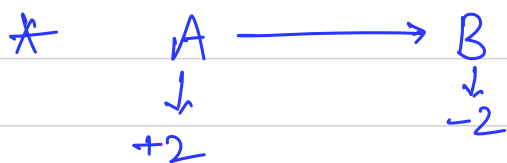
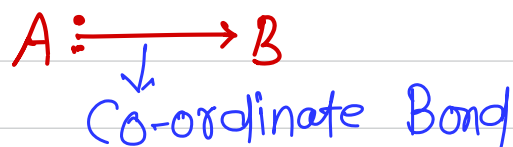
Case-1 In homoatomic Molecules:-



Case-2 - In Heteroatomic Molecules:
(if E.N of B > E.N of A)



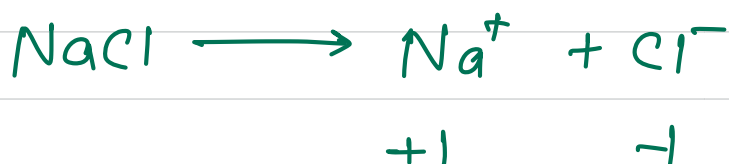
(B) In Case of Co-ordinate Bond (if E.N of B > E.N of A)



(C) In Case of Ionic Bond:-

O.S of Cation = Charge on Cation

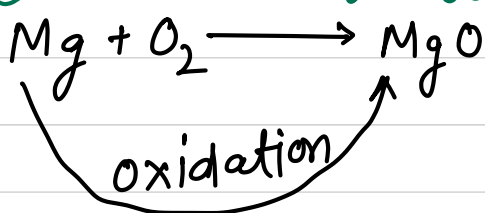
O.S of Anion = Charge on anion.



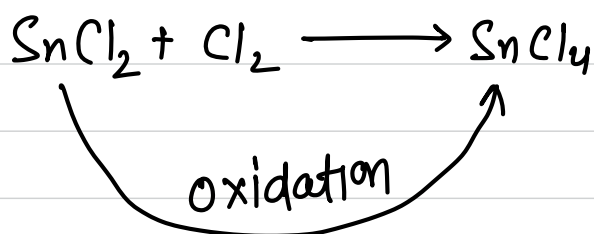
Oxidation and Reduction:-

(A) old Concept oxidation

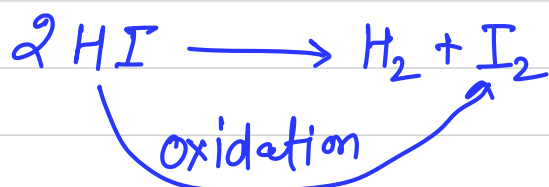
① Addition of oxygen



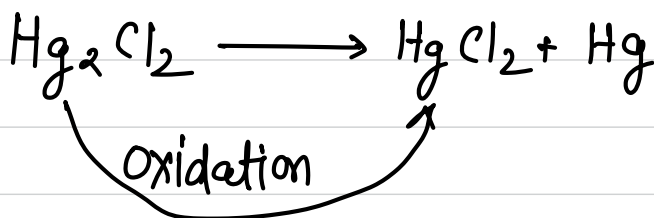
② Adding of more electronegative element



③ Removal of Hydrogen

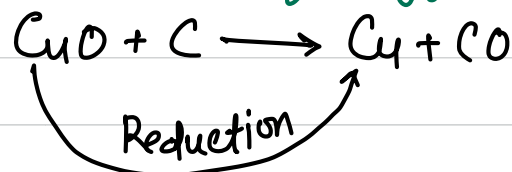


④ Removal of electropositive element



Reduction

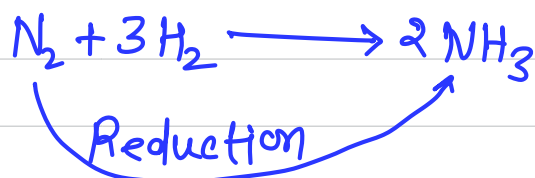
① Removal of oxygen



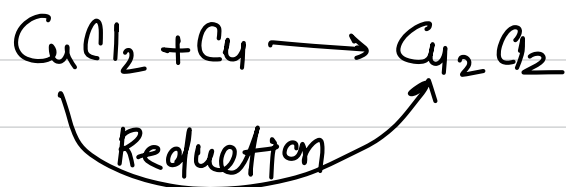
② Removal of more electro negative element



③ Addition of Hydrogen



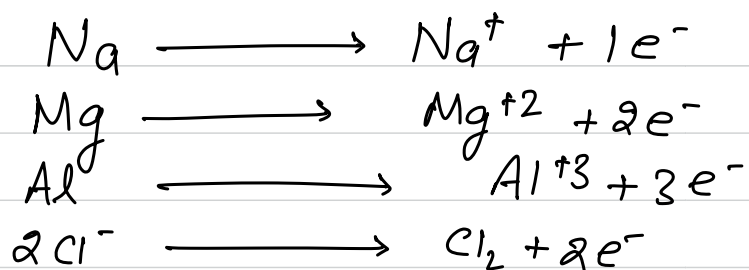
④ Addition of electropositive element



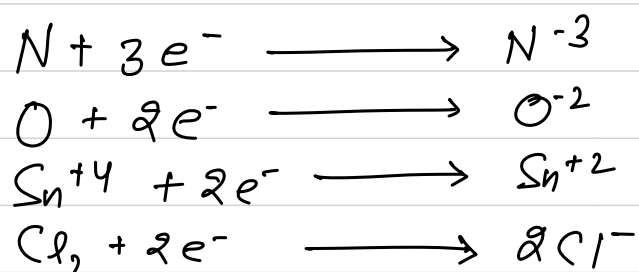
Modern Concept :-

(B) Modern Concept :-

Oxidation:- Loss of e^- is known as oxidation.



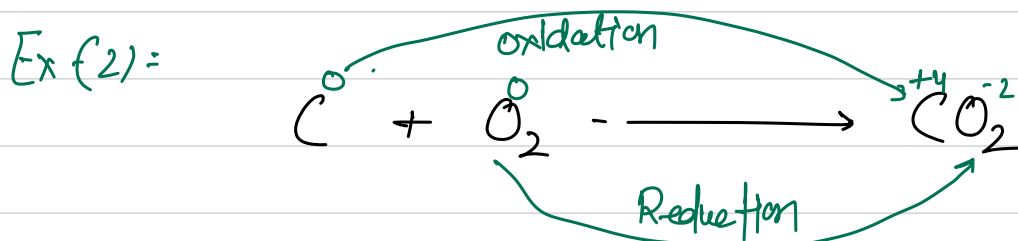
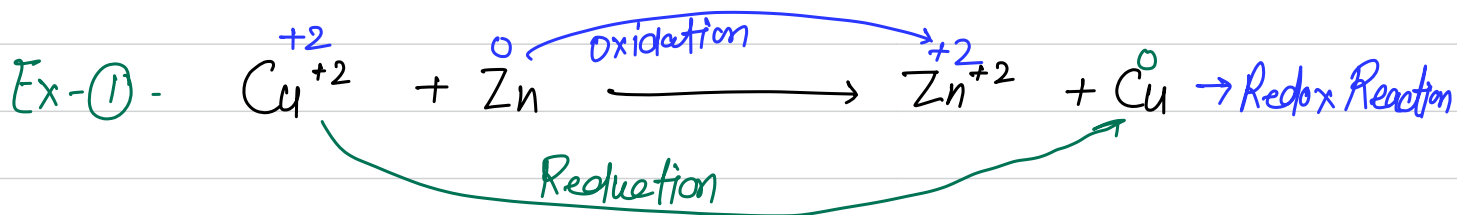
Reduction:- Gain of e^- is known as Reduction.



(c) Oxidation & Reduction is terms of O.N.

Oxidation:- If O.N of an element in a Reaction is increases is known as Oxidation.

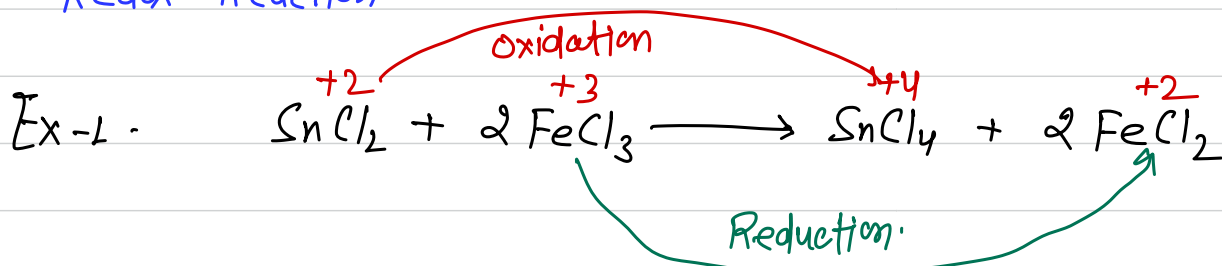
Reduction:- If O.N of an element in a reaction is decreases is known as Reduction.



Types of Redox Reaction:-

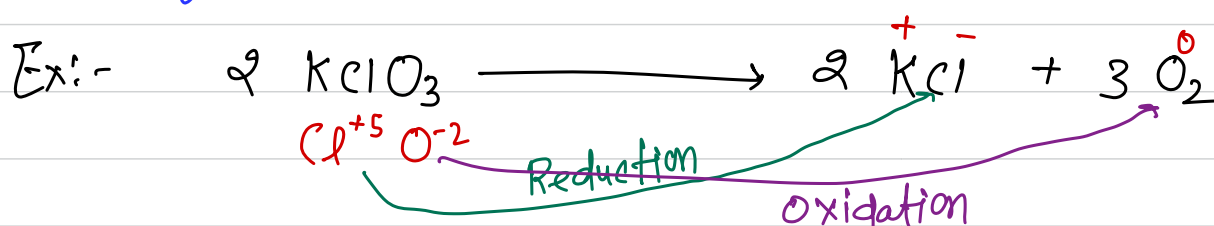
① Intermolecular Redox Reaction:-

When Oxidation and Reduction take place Separately in two different Species are known as intermolecular Redox Reaction.



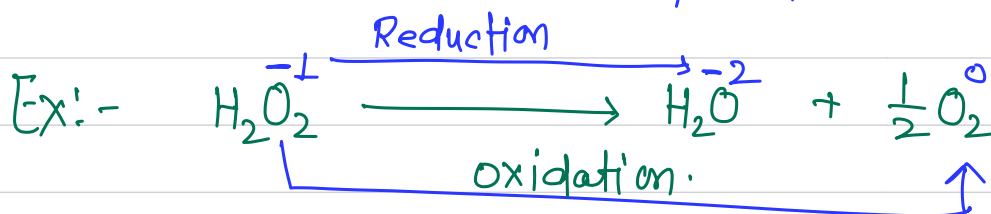
② Intramolecular Redox Reaction:-

When oxidation and Reduction take place in a Single Species is known as intramolecular R.R.



③ Disproportionation Redox Reaction:-

When Oxidation and Reduction take place of Same element in a Compound.

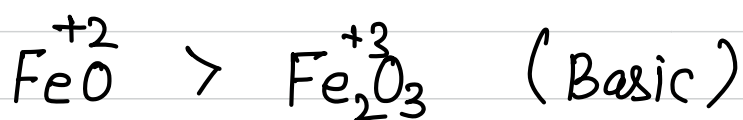


Application of Oxidation Number :-

① To Compare strength of an acid and base:-

Strength of Acid $\propto$ O.N

Strength of base $\propto \frac{1}{\text{O.N}}$



② To Find oxidising Agent (O.A) & Reducing Agent (R.A)

(a) - Oxidising Agent (O.A) / Oxidant:-

→ Substance which oxidise the other Substance and reduce itself are known as Oxidising Agent.

- It Reduce itself
- It is e^- acceptor

Some important O.A :

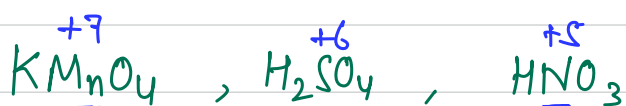
All high electronegative element like $\rightarrow \text{F, O, N, Cl}$

All Metal oxide like $\rightarrow \text{Na}_2\text{O, MgO, CaO, Al}_2\text{O}_3, \text{BaO}$

non-metal oxide like $\rightarrow \text{SO}_2, \text{CO}_2, \text{H}_2\text{O}_2 \dots$

Imp:

In a Compound or ion of an element is present in its highest oxidation state then it Can act as O.A.



(B) Reducing Agent (R.A) / Reductant:

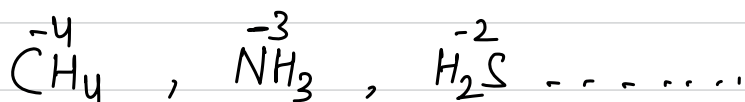
Substance which reduce the other Substance and oxidise itself is known as Reducing Agent.

- It oxidise itself
- It is e^- donor

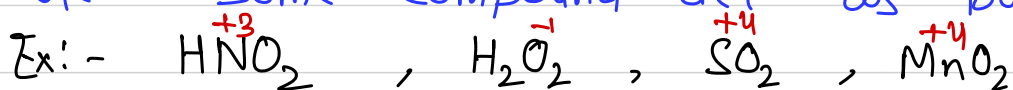
Some important R.A:-

- All Metals like $\rightarrow Na, K, Rb, Ba, Mg, \dots$
- All Metal hydride like $\rightarrow NaH, KH, CaH_2$
- All Hydroacids like $\rightarrow HCl, HBr, HI, HF, H_2S, \dots$

* In a Compound or ion of an element is Present in its lowest Oxidation state then it Can act as Reducing Agent.

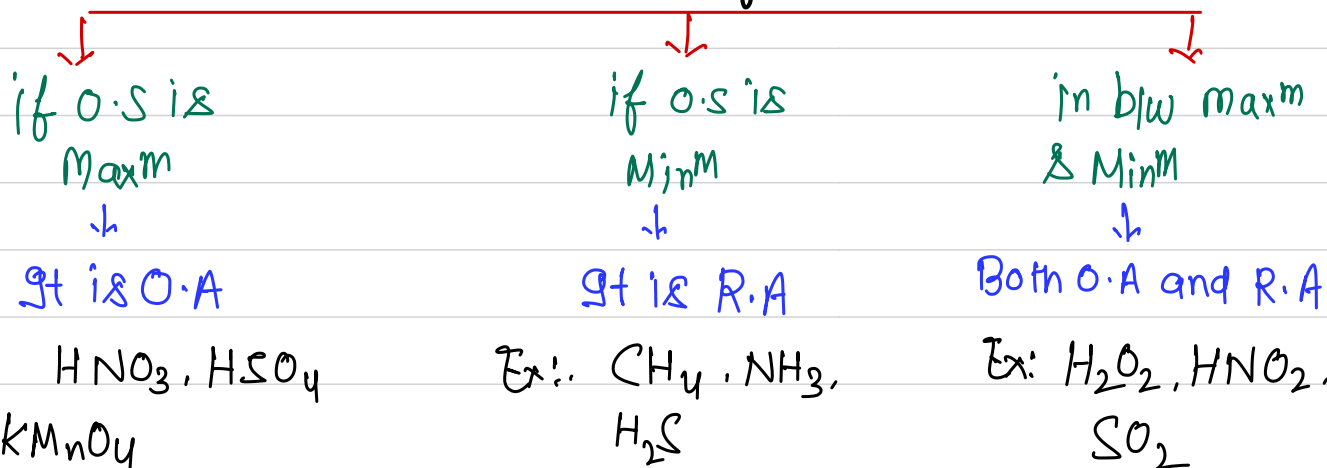


Note:- Some Compound act as Both O.A and R.A



Flow chart to Find O.A/R.A:-

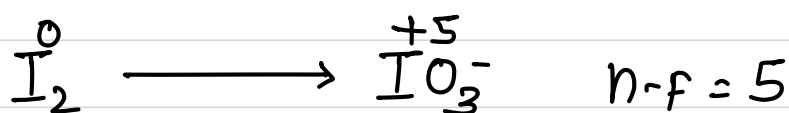
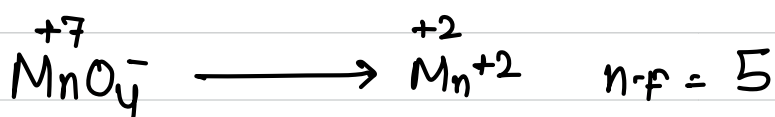
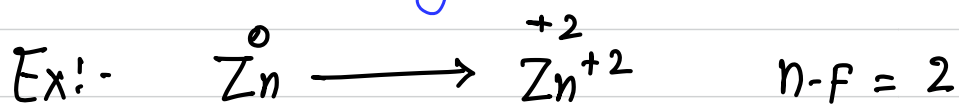
To Find O.S of Central Atom



④ Equivalent Weight of O.A and R.A : -

$$\text{Equivalent weight} = \frac{\text{Molar weight}}{n\text{-Factor}}$$

n-Factor: n-Factor for oxidant and Reducant is defined as total No. of mole of e^- gain and lost by one mole of Substance.



n-Factor Calculation in Disproportion Redox Rxn:-



$$n\text{-f} = \frac{n_1 \times n_2}{n_1 + n_2} \quad n_1 = 2 \quad n_2 = 1 \quad n_f = \frac{2 \times 1}{2 + 1} = \frac{2}{3}$$



$$n\text{-f} = \frac{2 \times 2}{2 + 2} = \frac{4}{4} = 1$$

$$\boxed{n_f = 1}$$

Some Important O.A & R.A with Eq. wt :-

S.No.	Species	Charged to	n-f	Eq. wt
1.	MnO_4^- (O.A)	Mn^{+2} (acidic Medium)	5	$\frac{M}{5}$
2.	MnO_4^- (O.A)	MnO_2 (Neutral/weak basic Medium)	3	$\frac{M}{3}$
③	$\text{Cr}_2\text{O}_7^{+6-2}$ (O.A)	Cr^{+3} (Acidic Medium)	6	$\frac{M}{6}$
④	Fe^{+2} (R.A)	Fe^{+3}	1	$\frac{M}{1}$
⑤	H_2O_2	H_2O^{+2}	2	$\frac{M}{2}$
⑥	I^- (R.A)	I_2^0 (acidic Medium)	1	$\frac{M}{1}$

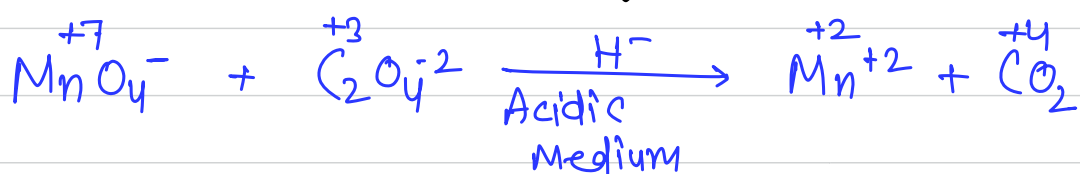
Law of Equivalent :-



$$g - \text{eq. A} = g - \text{eq. B} = g - \text{eq. C} = g - \text{eq. D}$$

$$a \times h_A = b \times h_B = c \times h_C = d \times h_D$$

Q → Find moles of Oxalate ion Oxidised by 1 mol of MnO_4^- Completely in Acidic Medium.



$$n-f = 5 \quad n_f = 2$$

$$g\text{-eq MnO}_4^- = g\text{-eq} \cdot \text{C}_2\text{O}_4^{2-}$$

$$\text{Mole} \times n-f = \text{mole} \times n_f$$

$$1 \times 5 = n \times 2$$

$$n = \frac{5}{2}$$

Balancing of Redox Reaction:-

Any chemical Rxn Can be balanced by Conserving of atom and charge.

(i) Oxidation Number Method

(ii)- Ion e^- method.

(A) Oxidation Number Method:-

Step-1 - Assign Oxidation No.

Step-2 - Write both Oxidation and Reduction Reaction Separately.

Step-3 - Balance the atom which are responsible for change in oxidation No.

Step-4 - Calculate change in Oxidation No.

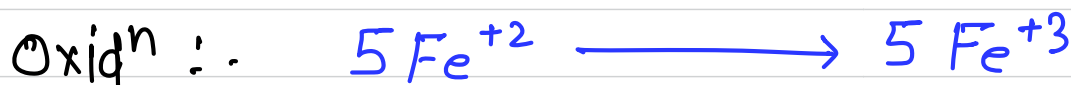
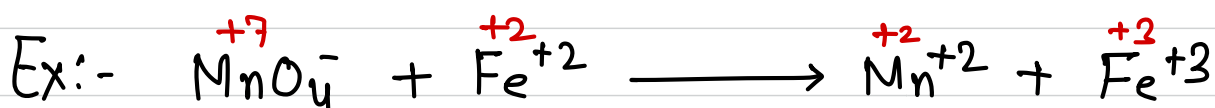
Step-6 - Now added both the Reaction.

Step-7 - If any reaction Component left initially then write in proper Side and balance them.

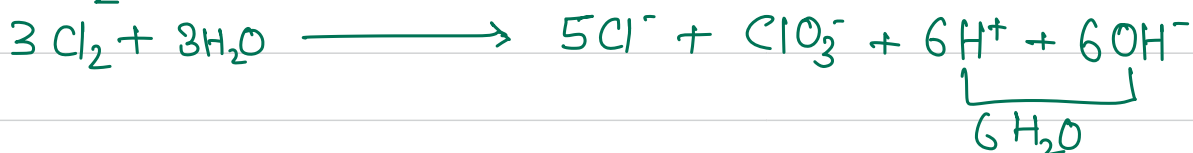
Step-8 - Balance oxygen('O') by adding $\rightarrow \text{H}_2\text{O}$

Balance hydrogen(H) by adding $\rightarrow \text{H}^+$

Step-9 - If Medium is basic then added OH^- ion equal to H^+ ion in both the Side.



Balancing of disproportionation R.R:-



(B) → Ion electron Method:-

Step-1 - Same as above

Step-2 - Same as above

Step-3 - Balance all atom except hydrogen and Oxygen.

Step-4 - Now balance Oxygen by adding $\rightarrow \text{H}_2\text{O}$
Balance hydrogen by adding $\rightarrow \text{H}^+$

Step-5 → Balance charge in both the Rxn by adding e^- .

Step-6 → Now balance e^- in both the Reaction.

Step-7 $\rightarrow$ Added both the Rxn:

Step-8 $\rightarrow$ if Medium is basic then added OH^- ion equal to H^+ ion in both the Side.

